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ORIGINAL ARTICLE

Effect of a State-Level Vaping Prevention Campaign on Beliefs and Behaviors in Young People

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ABSTRACT

Background: Vaping prevention media campaigns have promising effects on harm perceptions but have yet to demonstrate impacts on vaping behaviors in young people. The goal of this study was to evaluate the effect of Vermont's vaping prevention campaign (Unhyped) on vaping-related beliefs and behaviors.

Methods: Data come from Waves 5 (Winter 2020) and 9 (Winter 2021) (n=433) of the Policy and Communication Evaluation Vermont study, a longitudinal online cohort study of adolescents and young adults ages 12-25. Analyses examined associations between awareness of Unhyped in 2020 and outcomes in 2020 and 2021. Primary analyses compared participants aged 12-17 (campaign target) who reported awareness of Unhyped to propensity score-matched controls.

Results: In 2020, more adolescents aware of the Unhyped campaign perceived great risk from weekly vaping compared with matched controls (49.3% vs. 24.3%; p=0.019). Adolescents aware of Unhyped in 2020 were also less likely than matched controls to report willingness to try a vapor product in the next year (15.8% vs. 40.1%; p=0.048) or use one if offered by a friend (21.4% vs. 49.1%; p=0.031) in 2021. There was no relationship between brand awareness and vaping behaviors in 2020 or 2021.

Conclusions: Although there were no effects of awareness of Vermont's Unhyped campaign on vaping behaviors, the campaign was effective in altering short-term risk perceptions and reducing susceptibility to vape in adolescents one year later.

KEYWORDS
Youth; young adult; vaping; campaign

Introduction

Successful mass media education campaigns on tobacco have targeted knowledge and beliefs as the precursor to changing attitudes, and ultimately, tobacco use behavior (Centers for Disease Control and Prevention, 2014; Farrelly et al., 2009; Hershey et al., 2005; Kranzler et al., 2017; Zhao et al., 2016). These pathways have been empirically supported by evaluations of the truth campaign (Hersey et al., 2003; Hershey et al., 2005) and serve as the basis for the FDA's The Real Cost campaign (Duke et al., 2018; Duke et al., 2019; Farrelly et al., 2017; Kranzler et al., 2017; Zhao et al., 2016). Specifically, structural equation modeling in the truth campaign evaluation showed that negative beliefs about tobacco industry practices led to negative attitudes about the tobacco industry and subsequently, changes in smoking behavior in adolescents and young adults (AYA) (Hersey et al., 2003; Hershey et al., 2005). Preliminary evidence from FDA's The Real Cost campaign on vaping (U.S. Department of Health and Human Services) has shown similar effects on increasing campaign-targeted harm perceptions of and negative attitudes about e-cigarettes in adolescents (Kowitt et al., 2023; MacMonegle et al., 2024; MacMonegle et al., 2022; Rohde et al., 2021). In addition, a randomized trial found reduced susceptibility to vape with greater perceived message effectiveness among those exposed to The Real Cost campaign (Ma et al., 2024; Noar et al., 2022). Other e-cigarette health communication campaigns have been developed and tested, showing promise in increasing vaping-related harm perceptions among young people (England et al., 2021; Villanti et al., 2021). To date, there have been no published studies that demonstrate impacts of vaping prevention campaigns on vaping behaviors, either by reducing trial of electronic vaping products (EVPs) or frequency of vaping. However, the media landscape has evolved during the emergence of EVPs to include online platforms like social media, which present opportunities but also challenges for

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measuring exposure to campaigns and resulting behavioral changes (Ghahramani et al., 2022).

While the ideal campaign evaluation would include a randomized design to minimize bias in estimating the intervention's effect in a population, that is neither feasible nor ethical for most public health media campaigns. FDA's Fresh Empire campaign is one of the few to use a randomized design to evaluate 15 treatment markets receiving a targeted digital media campaign to prevent tobacco use in hip hop adolescents with 15 control markets; findings suggest that treatment markets reported higher awareness of Fresh Empire than control markets over time, and exposure was associated with increased agreement with five belief statements related to addiction/control, being a bad influence on family/friends, and cosmetic effects of smoking on breath and attractiveness (Guillory et al., 2022). Similarly, FDA's This Free Life campaign targets lesbian, gay, bisexual, and transgender (LGBT) young adults using a randomized design comparing 12 treatment and 12 control markets (Guillory et al., 2021). Evaluations find greater brand and ad awareness in treatment than control markets and small but significant campaign effects on five tobacco-related beliefs (Crankshaw et al., 2022).

More commonly, tobacco and substance use media campaign evaluations have consisted of observational studies with campaign exposure defined by market-level media delivery (gross rating points, GRPs, or target rating points, TRPs) and self-reported ad exposure (Davis et al., 2009; Davis et al., 2018; Duke et al., 2018; Duke et al., 2019; Farrelly et al., 2017; Vallone et al., 2011). A few evaluations have also used matching methods to approximate a randomized design using the available observational data (Lu et al., 2001; Villanti et al., 2011; Zanutto et al., 2005). These studies estimate the probability of assignment to the exposure using a number of observed covariates using logistic regression to generate a propensity score, then match or subclassify "exposed" or "unexposed" individuals based on similar covariates as summarized by the propensity score. Two studies using pilot data from a National Anti-Drug Media Campaign demonstrated no effect of the campaign on intentions to use marijuana or inhalants in propensity score matched samples of adolescents (Lu et al., 2001; Zanutto et al., 2005). Over 20 covariates were used for matching, including demographics, use of other substances, media consumption, and extracurricular activity involvement, and the findings were consistent with the findings of the formal evaluation (Hornik et al., 2008). A third study found no effect of a national smoking cessation mass media campaign (Become An EX) for adult cigarette smokers on quit attempts or cessation cognitions at six-month follow-up in the matched sample (Villanti et al., 2011). Ten covariates were used for matching, including demographics, quit attempts, and indicators of nicotine dependence; findings were inconsistent with the results of the original evaluation (Vallone et al., 2011). After excluding a subset of participants who had quit smoking at six-month follow-up, however, results were consistent between the matched and unmatched analyses and favored the intervention.

Application of matching methods may reduce bias in estimating the effect of large-scale prevention efforts but have not been widely used in digital media campaign evaluations. In response to state (Vermont Department of Health, 2020) and national increases in e-cigarette use (U.S. Department of Health and Human Services, 2016) and the importance of countering vape product marketing on social media (Lee et al., 2021), the Vermont Department of Health's Tobacco Control Program (VTCP) implemented a state-wide targeted digital media campaign for vaping prevention in adolescents ages 13-17 (Unhyped) in Spring 2019 (Williams, 2020). Using data collected during 2020-2021, the goal of the current analysis was to compare vaping-related harm perceptions and patterns of use among AYA aware of)Vermont's Unhyped e-cigarette prevention media campaign with a matched sample of unaware AYA. We hypothesized that AYA aware of the campaign would report more accurate campaign-targeted beliefs and harm perceptions and lower nicotine EVP use than those unaware.

Materials and methods

The implementation of Unhyped coincided with the launch of the Policy and Communication Evaluation (PACE) Vermont Study, an online cohort study of 1,500 Vermont AYA recruited *via* digital and print ads, dissemination among prevention networks, and participant referrals (Villanti et al., 2020). At the time of enrollment, all participants provided consent electronically; adolescent participation required parental consent in an online form sent directly to a parent contact prior to providing assent in their own form. This research was approved by the University of Vermont Institutional Review Board and received a Certificate of Confidentiality from the National Institutes of Health over a total of nine survey waves. Data from two waves conducted one year apart were used in these analysis.

Wave 5 took place from December 8, 2020 - January 4, 2021 and included 827 participants (221 adolescents ages 12-17, 606 young adults ages 18-25). Wave 9 was collected from November 19 - December 20, 2021 and included 524 participants (145 adolescents, 379 young adults) representing 63% follow-up from the Wave 5 sample. Data for the analyses were limited to 433 participants at Wave 5 with follow-up data at Wave 9 (127 adolescents, 306 young adults) who had complete data on awareness of the state's Unhyped media campaign. Waves 5 and 9 were used for these analyses to examine the prospective effects of the Unhyped campaign over one-year at our most recent wave. Bivariate tests showed no significant differences between those who remained in the study and those who were lost to follow-up at Wave 9. Ongoing oversight of study analyses are provided by the Rutgers Institutional Review Board (#2022000166).

Unhyped campaign

In December 2018, the VTCP hosted focus groups at two Vermont high schools to test educational messaging and learn more about attitudes and behaviors associated with vaping. Responses from 24 teen participants showed that facts on the chemicals found in vapes, along with the subsequent health consequences, were considered attention-grabbing and convincing. Additionally, vape product use was identified as a social activity among teens. To reach the primary audience of teens aged 13-17 who currently vape or were susceptible to future use with the goal of increasing awareness of the harms associated with vaping, VTCP collaborated with its counter-marketing contractor, Rescue Agency, to create the Unhyped digital media campaign (www.UnhypedVt.com) to increase knowledge and correct perceptions, attitudes and beliefs about vaping. At the recommendation of Rescue, digital advertising was selected as an intervention strategy for its cost efficiency and targeting capability (Centers for Disease Control and Prevention, 2014).

VTCP launched Unhyped's first message package, "See Through," in March 2019 (details on campaign packages and timing are presented in Supplemental Table 1). The package included programmatic pre-roll video, an ad served before the user's chosen web content began, along with social media promotions (Facebook, Instagram and YouTube) that engaged teens aged 13-17 with vaping messages on channels through which they were already consuming content, targeting by geography (urban and rural counties) and age, along with an added layer of targeting to reach those who were watching vape influencers and vape-specific content on YouTube.

During the 2019 legislative session, three bills were passed to reduce youth access and tobacco use, specifically vaping, which went into effect in summer and fall, one of which was raising the minimum age to purchase tobacco to 21 (effective September 2019). Consequently, the VTCP expanded its target audience to Vermont AYAs ages 13—20 years at risk of vaping, running the second message package, "Flash Drive" from July 29—September 30, 2019 with the key message about the addictiveness of vaping, and the third message package, "Deceptions on Display" from March 20—April 30, 2020 with the key message about the vape industry's deceptive marketing. The VTCP ran the next Unhyped message package "Defenseless" from September 22, 2020—November 13, 2020, weeks prior to the PACE Vermont Wave 5 survey. The key message was on the health consequences of vaping, including respiratory illness.

Two message packages were delivered in 2021: "Hidden Weakness" messaged to those ages 13—20 about chemicals in vapes that can damage the lungs from March 4, 2021—April 30, 2021, and "Great Manipulator" targeted teens aged 13—17 with messages on vaping interfering with mental health and ran from December 14, 2021—February 28, 2022; "Great Manipulator" was in market during the last two weeks of the Wave 9 data collection.

Brand awareness

Brand awareness was defined as awareness of Vermont's Unhyped campaign (Evans et al., 2015; Evans et al., 2018; Rath et al., 2021; Vallone et al., 2017). Participants were presented with the Unhyped logo and asked "Have you ever seen or heard information about UNHYPED?" with response options "yes," "no," and "I don't know." Those reporting "yes" were defined as aware of Unhyped and asked to describe their frequency of exposure to Unhyped messages in the past 3 months, how much they liked Unhyped, and which phrases they remembered from Unhyped: (a) Vapes deliver more nicotine to your lungs than smoking (Wave 5 only), (b) Vape clouds contain as many as 31 different chemicals, including toxic metals that can build up in your lungs over time, (c) Toxic chemicals, like formaldehyde and acetone, can get in your lungs when you vape, (d) Vapes that contain nicotine rewire your brain for addiction, just like cigarettes do, (e) Vape companies created deceptive marketing campaigns and programs specifically for teens, including going into classrooms and stating their products are safe when they really aren't, and (f) Chemicals in vapes damage lungs on a cellular level, making you more vulnerable to viruses (Wave 9 only). These phrases addressed recall of content provided in the campaign (Supplemental Table 1).

Outcomes

Outcomes assessed included a campaign-targeted belief, perceived harm of EVP use, and EVP use/susceptibility.

Campaign-targeted belief

One item asked participants to assess the following belief targeted by the campaign at Wave 9: "One 5% vape pod can contain as much nicotine as an entire pack of cigarettes" with the response choices as "true," "false," and "don't know"

Perceived harm of EVP use

Perceived harm of vaping was assessed at both waves using an item adapted from the 2019 Vermont Youth Risk Behavior Survey: "How much do you think people risk harming themselves (physically or in other ways) if they use electronic vapor products weekly?" Responses were "No risk," "Slight risk," "Moderate risk," and "Great risk".

EVP use and susceptibility

Each wave of the PACE Vermont Study included measures on ever and past 30-day substance use, including cigarette, EVP, alcohol, and marijuana use (Villanti et al., 2020). Frequency of EVP use was asked at each wave, along with the number of occasions that participants had vaped nicotine, just flavoring, or marijuana in their lifetime, with categorical options ranging from "0 occasions" to "40 or more occasions." Among ever EVP users, an additional item captured past-year attempts to quit or cut down on vaping; participants could select all that apply from the following options - "yes, I tried to quit vaping," "yes, I tried to cut down on my vaping", "I don't know"—or a single response choice, "no." At Wave 9, participants were also asked three items assessing susceptibility of use EVP: "Do you think that you will try an electronic vapor product soon?", "Do you think that you will use an electronic vapor product in the next year?", and "If one of your best friends were to offer you an electronic vapor product, would you use it?" Participants were susceptible if they responded, "Definitely yes," "Probably yes," or "Probably not" vs. "Definitely not" (Pierce et al., 1996).

Covariates used in matching

Covariates used in propensity score matching models were collected at Wave 4 or Wave 5. For the full AYA sample (presented in Supplemental Table 2), covariates included age (continuous); sex (male/female); race/ethnicity (non-Hispanic white or other); employment status; ever use of cigarettes, EVPs with nicotine, alcohol, and marijuana; and past 12-month awareness of national vaping prevention campaigns. Employment was categorized as working full-time (at least 35 h per week), part-time (15-34 h per week), part-time (<15 h per week), or not currently working. Awareness of national campaigns was defined by reporting being aware of either the truth campaign or The Real Cost campaign in the past year. Given a different distribution of characteristics among the sub-sample of adolescents aged 12-17, covariates used in propensity score matching models for adolescents included sex; ever use of EVPs, alcohol, and marijuana; and past 12-month national campaign awareness.

Statistical analyses and data exploration

Propensity score matching was conducted in the Wave 5 data using the 'matchit' package in R Studio (2023.06.2 + 561). Matching minimized the standardized mean difference in the logits of propensity scores between the aware and unaware groups (Supplemental Figures 1 and 2). Caliper width (0.25) was standard across models, as was utilizing a generalized linear model to estimate the propensity score. Methods of matching included: (a) optimal full matching, (b) matching with replacement, and (c) optimal variable ratio matching. Optimal full matching maximized the number of unaware to aware participants, resulting in no umatched control units, and created the most balanced groups (Rosenbaum, 2020; Stuart, 2010; Stuart & Green, 2008).

For the primary analysis, we focused on the 127 adolescents (ages 12-17 years) with data at Waves 5 and 9 because this age group was initially targeted by the Unhyped campaign. In this group, we conducted several bivariate analyses comparing the aware participants ($n = 67$) with a) the unaware participants ($n = 60$), and b) the propensity score matched unaware controls ($n = 60$), using Pearson's chi-square tests. First we compared the baseline characteristics at Wave 5 (2020). Next, we examined the association between awareness of Unhyped and vaping-related beliefs and behaviors within Wave 5. Then we used prospective bivariate analyses to test the association between Unhyped awareness at Wave 5 (2020) with vaping-related beliefs and behaviors one year later (Wave 9, 2021). We also compared participant characteristics and vaping-related beliefs and behaviors in the matched group of aware participants ($n = 168$) to unaware and matched controls among the full sample of AYA (supplemental materials). Finally, we explored participant perceptions of the Unhyped campaign and perceptions of vaping harms among those aware of Unhyped at Waves 5 and 9.

Results

Among adolescent (12-17 years) participants in the PACE Vermont Study at Wave 5 with follow-up data at Wave 9 ($n = 127$), 66% were female, 21% had ever used EVPs, 34% had ever used alcohol, 17% had ever used marijuana, and 74% had seen other national vaping prevention ads in the past year (Table 1). Awareness of the Unhyped campaign declined from 53% ($n = 67$) at Wave 5 to 44% ($n = 56$) at Wave 9. In the full sample of adolescents at Wave 5, those aware of Unhyped were more likely to be male and to be aware of national vaping prevention campaigns. There were no differences between the aware and matched control groups.

In the full sample of AYAs ($n = 433$), 71% were young adults, and substance use was higher than among adolescents (Supplemental Table 2). AYAs aware of Unhyped were younger, more likely to be male and to be unemployed than those unaware; aware participants were also less likely to report ever use of cigarettes, alcohol, or marijuana than unaware participants (Supplemental Table 2). Matching improved balance across the aware and unaware participants, removing differences between the groups.

Table 1. Baseline characteristics of adolescent (12-17 years) participants aware and unaware (unmatched and matched samples) of Unhyped at Wave 5 (December 2020 – January 2021), PACE Vermont Study.

	All ($n = 127$) n (%)	Aware[a] ($n = 67$; 52.8%) n (%)	Unaware ($n = 60$; 47.2%) n (%)	p	Matched controls ($n = 60$) Weighted n (%)	p
Sex at birth				0.034		0.900
Female	82 (65.6)	37 (56.9)	45 (75.0)		34 (56.2)	
Male	43 (34.4)	28 (43.1)	15 (25.0)		26 (43.7)	
Ever used EVPs				0.446		0.758
No/IDK	100 (78.7)	51 (76.1)	49 (81.7)		48 (79.3)	
Yes	27 (21.3)	16 (23.9)	11 (18.3)		12 (20.7)	
Ever used alcohol				0.906		1.000
No/IDK	84 (66.1)	44 (65.7)	40 (66.7)		39 (65.6)	
Yes	43 (33.9)	23 (34.3)	20 (33.3)		21 (34.4)	
Ever used marijuana				0.853		0.873
No/IDK	105 (82.7)	55 (82.1)	50 (83.3)		50 (84.0)	
Yes	22 (17.3)	12 (17.9)	10 (16.7)		10 (16.0)	
Past 12-month awareness of the truth or The Real Cost campaigns				0.003		0.847
No/Not Sure	33 (26.0)	10 (14.9)	23 (38.3)		10 (17.2)	
Yes	94 (74.0)	57 (85.1)	37 (61.7)		50 (82.8)	

[a]n = 3 aware participants excluded from matched analyses at Wave 5.
Note. Categorical variables were tested using Pearson's chi-square test.

Associations between Unhyped awareness and vaping-related beliefs and behaviors in Winter 2020

The unmatched and propensity-score matched associations between awareness of Unhyped and vaping-related beliefs and behaviors are presented in Table 2 (adolescents) and Supplemental Table 3 (AYA). At Wave 5, more adolescents aware of the Unhyped campaign perceived great risk of weekly vaping compared with those unaware (49.3% vs. 24.3% matched controls; $p = 0.019$). There were no other associations between awareness of Unhyped and any of the Wave 5 outcomes among adolescents. Among AYA combined, unmatched and matched analyses showed no correlation between awareness of Unhyped and any of the outcomes at Wave 5 (Supplemental Table 3).

Prospective associations between Unhyped awareness and vaping-related beliefs and behaviors at one-year follow-up

Table 3 presents the prospective associations between Unhyped awareness at Wave 5 and vaping-related outcomes at Wave 9 in the matched and unmatched samples of adolescents. In the matched analyses, among adolescents who had not used EVPs in the past 30 days, those aware at Wave 5 were less susceptible to EVP use than those unaware, less likely to report willingness to try a vapor product in the next year (15.8% vs. 40.1% matched controls; $p = 0.048$) or use one if offered by a friend (21.4% vs. 49.1% matched controls; $p = 0.031$) one year later. Among the full sample of AYA, unmatched and matched analyses showed no correlation between awareness of Unhyped at Wave 5 and any of the outcomes at Wave 9 (Supplemental Table 4).

Perceptions of Unhyped among aware AYAs

Among adolescent participants aware of Unhyped at Wave 5 ($n = 67$) and Wave 9 ($n = 56$), the average number of times adolescents reported seeing or hearing Unhyped in the past three months was 3.6 at Wave 5 and 2.9 at Wave 9 (Table 4). Most adolescents who were aware of Unhyped were either neutral (neither liked nor disliked; Wave 5: 55%, Wave 9: 61%) or liked (Wave 5: 21%, Wave 9: 23%) the campaign. At both waves, a majority of adolescents recalled campaign-targeted vaping-related facts, except for "Vape companies created deceptive marketing campaigns and programs specifically for teens, including going into classrooms and stating their products are safe when they really aren't," recalled by 45% of adolsecents at Wave 5 and 30% at Wave 9, as well as "Vape chemicals damage lungs, making you more vulnerable to viruses," which was recalled by only 30% of adolescents at Wave 9.

Table 2. Associations between Unhyped awareness and adolescent vaping-related beliefs and behaviors at Wave 5 (December 2020 – January 2021), in unmatched and matched samples, PACE Vermont Study.

	All ($n = 127$) n (%)	Aware[a] ($n = 67$; 52.8%) n (%)	Unaware ($n = 60$; 47.2%) n (%)	p	Matched controls ($n = 60$) Weighted n (%)	p
Vaping-related belief						
Risk of harm from using EVPs weekly				0.040		0.019
No or slight risk	24 (18.9)	14 (20.9)	10 (16.7)		9 (15.7)	
Moderate risk	51 (40.2)	20 (29.9)	31 (51.7)		36 (60.0)	
Great risk	52 (40.9)	33 (49.3)	19 (31.7)		15 (24.3)	
How much people harm themselves using EVPs				0.699		0.743
No or a little harm	21 (16.5)	12 (17.9)	9 (15.0)		9 (14.2)	
Some harm	44 (34.6)	21 (31.3)	23 (38.3)		17 (28.5)	
A lot of harm	62 (48.8)	34 (50.7)	28 (46.7)		34 (57.4)	
Relative harm of EVPs compared with cigarettes				0.062		0.141
Less harmful	21 (16.5)	9 (13.4)	12 (20.0)		11 (17.7)	
About the same	69 (54.3)	43 (64.2)	26 (43.3)		27 (45.5)	
More harmful	37 (29.1)	15 (22.4)	22 (36.7)		22 (36.8)	
Relative harm of vaping nicotine compared with marijuana				0.958		0.371
Less harmful	11 (8.7)	6 (9.0)	5 (8.3)		11 (19.1)	
About the same	66 (52.0)	34 (50.7)	32 (53.3)		25 (42.5)	
More harmful	50 (39.4)	27 (40.3)	23 (38.3)		24 (38.4)	
Vaping-related behaviors						
Lifetime vaping of nicotine				0.827		0.513
0	113 (89.0)	60 (89.6)	53 (88.3)		51 (84.4)	
1+	14 (11.0)	7 (10.4)	7 (11.7)		9 (15.6)	
Lifetime vaping of marijuana				0.360		0.617
0	113 (89.0)	58 (86.6)	55 (91.7)		54 (90.6)	
1+	14 (11.0)	9 (13.4)	5 (8.3)		6 (9.4)	
Lifetime vaping of flavoring				0.226		0.222
0	118 (92.9)	64 (95.5)	54 (90.0)		53 (89.1)	
1+	9 (7.1)	3 (4.5)	6 (10.0)		7 (10.9)	
Lifetime vaping of CBD				0.911		1.000
0	123 (96.9)	65 (97.0)	58 (96.7)		58 (96.9)	
1+	4 (3.1)	2 (3.0)	2 (3.3)		2 (3.1)	
Used EVPs within past 30 days	7 (5.5)	4 (6.0)	3 (5.0)	0.811	3 (4.7)	0.706

[a]n = 3 aware participants excluded from matched analyses at Wave 5.
Note. Categorical variables were tested using Pearson's chi-square test unless otherwise noted.

Table 3. Prospective associations between Unhyped awareness at Wave 5 (December 2020 – Janaury 2021) and adolescent vaping-related beliefs and behaviors at Wave 9 (November-December 2021) in unmatched and matched samples, PACE Vermont Study.

	All (*n* = 127) n (%)	Aware[a] (*n* = 67; 52.8%) n (%)	Unaware (*n* = 60; 47.2%) n (%)	*p*	Matched controls (*n* = 60) Weighted n (%)	*p*
Vaping-related beliefs						
One 5% vape pod can contain as much nicotine as an entire pack of cigarettes.				0.215		0.079
False/Don't Know	52 (40.9)	24 (35.8)	28 (46.7)		34 (57.5)	
True	75 (59.1)	43 (64.2)	32 (53.3)		26 (42.5)	
Risk of harm from using EVPs weekly				0.273		0.117
No or slight risk	26 (20.5)	13 (19.4)	13 (21.7)		10 (16.5)	
Moderate risk	56 (44.1)	26 (38.8)	30 (50.0)		36 (59.4)	
Great risk	45 (35.4)	28 (41.8)	17 (28.3)		14 (24.0)	
Relative harm of EVPs compared with cigarettes				0.736		0.500
Less harmful	24 (18.9)	12 (17.9)	12 (20.0)		18 (29.8)	
About the same	66 (52.0)	37 (55.2)	29 (48.3)		31 (51.5)	
More harmful	37 (29.1)	18 (26.9)	19 (31.7)		11 (18.7)	
Relative harm of vaping nicotine compared with marijuana				0.964		0.154
Less harmful	8 (6.3)	4 (6.0)	4 (6.7)		11 (18.7)	
About the same	54 (42.5)	28 (41.8)	26 (43.3)		19 (31.3)	
More harmful	65 (51.2)	35 (52.2)	30 (50.0)		30 (49.9)	
Change from Wave 5 in harm perceptions of using EVPs weekly						
Decrease	30 (23.6)	15 (22.4)	15 (25.0)	0.895	10 (16.9)	0.612
No change	74 (58.3)	39 (58.2)	35 (58.3)		41 (68.8)	
Increase	23 (18.1)	13 (19.4)	10 (16.7)		9 (14.3)	
Vaping-related behaviors						
Lifetime vaping of nicotine				0.604		0.924
0	100 (79.4)	52 (77.6)	48 (81.4)		47 (79.0)	
1+	26 (20.6)	15 (22.4)	11 (18.6)		12 (21.0)	
Lifetime vaping of marijuana				0.689		0.597
0	104 (81.9)	54 (80.6)	50 (83.3)		51 (85.2)	
1+	23 (18.1)	13 (19.4)	10 (16.7)		9 (14.8)	
Lifetime vaping of flavorings						
0	111 (87.4)	59 (88.1)	52 (86.7)	0.813	52 (85.9)	0.814
1+	16 (12.6)	8 (11.9)	8 (13.3)		8 (14.1)	
Lifetime vaping of CBD				0.386		0.316
0	118 (92.9)	61 (91.0)	57 (95.0)		57 (95.3)	
1+	9 (7.1)	6 (9.0)	3 (5.0)		3 (4.7)	
Ever use of EVPs						
Never used	92 (72.4)	44 (65.7)	48 (80.0)	0.122	47 (77.7)	0.266
Used at start of study	7 (5.5)	5 (7.5)	2 (3.3)		2 (3.1)	
Used for first time at Wave 5	20 (15.7)	11 (16.4)	9 (15.0)		10 (17.6)	
Used for first time at Wave 9	8 (6.3)	7 (10.4)	1 (1.7)		1 (1.6)	
Past 30-Day Use Across Waves				0.961		0.969
No Past 30-Day Use	108 (85.0)	56 (83.6)	52 (86.7)		51 (84.4)	
No Use at Wave 5, Use at Wave 9	12 (9.4)	7 (10.4)	5 (8.3)		6 (10.9)	
Use at Wave 5, No Use at Wave 9	5 (3.9)	3 (4.5)	2 (3.3)		2 (3.1)	
Use at Both Wave 5 and 9	2 (1.6)	1 (1.5)	1 (1.7)		1 (1.6)	
Used EVPs within past 30 days	17 (13.4)	10 (14.9)	7 (11.7)	0.590	8 (14.1)	1.000
No. days used EVPs within past 30 days ($M \pm SD$)[b]	12.2 (11.4)	12.7 (10.9)	11.6 (12.9)	0.848	9.2 (11.2)	0.481
Tried to quit or cut down EVP use in past year[c]	7 (20.6)	4 (17.4)	3 (27.3)	0.505	3 (22.6)	0.813
Vaping Susceptibility/Curiosity						
Susceptible to EVP use[d,e]	31 (28.4)	14 (24.6)	17 (32.7)	0.347	25 (49.4)	0.058
Try an EVP soon	18 (16.5)	8 (14.0)	10 (19.2)	0.466	19 (37.5)	0.053
Try an EVP in next year	23 (21.1)	9 (15.8)	14 (26.9)	0.155	20 (40.1)	0.048
Use EVP if offered by friend	27 (25.0)	12 (21.4)	15 (28.9)	0.374	25 (49.1)	0.031
Ever been curious about using an EVP[d,f]	39 (35.8)	18 (31.6)	21 (40.4)	0.338	26 (51.7)	0.107

[a]n = 3 aware participants excluded from matched analyses at Wave 5.
Note. Categorical variables were tested using Pearson's chi-square test. No. days used EVPs within past 30 days was tested using Wilcoxon Rank Sum test.
[b]Among those who used EVPs in the past 30 days.
[c]Among those who have ever used EVPs.
[d]Among those who have not used EVPs in the past 30 days.
[e]Respondents were not susceptible if they responded "Definitely Not" to "Do you think that you will try an electronic vapor product soon?", "Do you think that you will use an electronic vapor product in the next year?", and "If one of your best friends were to offer you an electronic vapor product, would you use it?".
[f]Responded" Definitely Yes", "Probably Yes", or "Probably Not" (vs." Definitely Not").

Almost all adolescents reported believing that the Unhyped campaign does not promote new vape flavors and does not recommend vaping over smoking. Most reported that Unhyped either moderately (Wave 5: 27%, Wave 9: 38%) or extremely (Wave 5: 53%, Wave 9: 47%) reveals facts about vaping. Adolescent participants were fairly split on whether they believed that Unhyped educates on Vermont tobacco law/policy. At Wave 5, 30% of adolescents aware of Unhyped reported having taken action as a result of information in Unhyped; only 11% reporting taking action at Wave 9. Most of those taking action decided not to vape (Wave 5: 24% of adolescents, Wave 9: 4%) or advised a

Table 4. Perceptions of Unhyped and vaping harms among adolescents aware of Unhyped at Wave 5 (December 2020 – January 2021) and Wave 9 (November-December 2021), PACE Vermont Study.

	Wave 5 (*n* = 67; 52.8%)	Wave 9 (*n* = 56; 44.1%)[a]
Number of times seen or heard Unhyped (mean, SD)	3.6 (1.8)	2.9 (1.6)
How much like Unhyped		
Really like	8 (11.9)	5 (8.9)
Like	14 (20.9)	13 (23.2)
Neither like nor dislike	37 (55.2)	34 (60.7)
Don't like	4 (6.0)	2 (3.6)
Really don't like	4 (6.0)	2 (3.6)
Vapes deliver more nicotine than smoking	34 (50.8)	–
Vape clouds contain 31 chemicals	40 (59.7)	35 (62.5)
Toxic chemicals in lungs when you vape	42 (62.7)	31 (55.4)
Vapes rewire your brain	41 (61.2)	30 (53.6)
Vape companies create deceptive marketing	30 (44.8)	17 (30.4)
Vape chemicals damage lungs, making you more vulnerable to viruses	–	27 (30.4)
Unhyped promotes new vape flavors		
Not at all	60 (93.8)	52 (94.6)
Slightly	1 (1.6)	2 (3.6)
Somewhat	2 (3.1)	1 (1.8)
Moderately	1 (1.6)	0 (0.0)
Extremely	0 (0.0)	0 (0.0)
Unhyped recommends vaping over smoking		
Not at all	60 (93.8)	54 (98.2)
Slightly	1 (1.6)	0 (0.0)
Somewhat	3 (4.7)	1 (1.8)
Moderately	0 (0.0)	0 (0.0)
Extremely	0 (0.0)	0 (0.0)
Unhyped reveals facts about vaping		
Not at all	5 (7.8)	1 (1.8)
Slightly	0 (0.0)	1 (1.8)
Somewhat	8 (12.5)	6 (10.9)
Moderately	17 (26.6)	21 (38.2)
Extremely	34 (53.1)	26 (47.3)
Unhyped educates on Vermont tobacco law/policy		
Not at all	15 (23.4)	13 (23.6)
Slightly	11 (17.2)	15 (27.3)
Somewhat	13 (20.3)	12 (21.8)
Moderately	16 (25.0)	7 (12.7)
Extremely	9 (14.1)	8 (14.6)
Taken action as a result of information in Unhyped	20 (29.9)	6 (10.7)
Action: Decided not to vape	16 (23.9)	2 (3.6)
Action: Decided to stop vaping	1 (1.5)	0 (0.0)
Action: Decided to reduce vaping	0 (0.0)	0 (0.0)
Action: Advised friend not to vape	5 (7.5)	3 (5.4)

[a]n = 25 participants at Wave 9 reported no awareness of the Unhyped campaign but endorsed awareness at Wave 5 (recanted).

friend not to vape (Wave 5: 8%, Wave 9: 5%). Among the full sample of AYA, recall of vaping-related facts targeted in the campaign was generally lower than among only adolescents (ranging from 31% to 52%) (Supplemental Table 5).

Discussion

The current study documents the rapid impact of a state-level prevention media campaign on vaping-related beliefs and prospective impact on vaping use susceptibility in adolescents ages 12-17. Awareness of Unhyped in 2020 was prospectively associated with risk perceptions of weekly harm of EVP use and negatively associated with a willingness to try an EVP in the next year or if offered by a friend asssessed in 2021. The harm perception finding was consistent in unmatched analyses, as well as those conducted in the propensity score-matched dataset where differences in characteristics of those aware and unaware of the campaign had been controlled.

Vaping susceptibility-related findings emerged after groups were matched, suggesting the effectiveness of the campaign among groups of adolescents who differed by brand awareness status but were similar on other characteristics. Consistent with our hypotheses, awareness of Unhyped was associated with campaign-targeted harm perceptions, but not with other targeted beliefs in matched analyses. Our findings align with other e-cigarette health communication campaigns that increased vaping-related harm perceptions, a precursor to behavior change, among young people (England et al., 2021; Villanti et al., 2021). Despite finding no relationship between brand awareness and EVP use, reduced susceptibility among aware participants suggests support for our hypotheses, and is consistent with evaluation of The Real Cost campaign (Ma et al., 2024; Noar et al., 2022).

Imbalances in brand awareness by sociodemographic characteristics highlight brand awareness among younger, male Vermonters. Given the higher prevalence of EVP use among Vermont female high school students in 2021 (Vermont Department of Health, 2021a), greater awareness in males compared to females may also reflect message reactance or message avoidance given that the messages directly countered EVP use (Hall et al., 2018). Of particular relevance, those aware of Unhyped reported lower ever use of cigarettes, alcohol, or marijuana than those unaware, supporting that campaign ads reached an at-risk or low-risk sample of teens and young adults. More of those aware of Unhyped also reported awareness of the truth campaign and/or FDA's The Real Cost campaign. Nevertheless, almost two-thirds of the unaware group had seen ads from national campaigns as well. A lack of effects of Unhyped on campaign-targeted beliefs could be due to a saturation of similar messaging received across campaigns. In addition, a waning of Unhyped campaign content over the course of our study period (during 2021) may have contributed to null effects at follow-up, which is reflected in the lower level of awareness and number of times having seen the campaign among study participants in Winter 2021. Although the campaign was expanded to reach young adults up to 20 years of age in 2019, more adolescents reported awareness; this may explain why effects were only found among adolescents but not among AYA when combined. Another explanation may be the difficulty of impacting vaping-related beliefs and behaviors in young adults (Villanti et al., 2022), despite positive response to vapring prevention messages (Villanti et al., 2021).

Strengths of the current study include the state-wide sample of AYAs with longitudinal follow-up, use of real-time survey data collection for evaluation of the state's vaping prevention mass media campaign, and application of matching methods to reduce bias in estimates of the campaign's effect on vaping-related beliefs and behaviors. Limitations include the focus on a single state, the convenience sample of Vermont AYAs, the concurrent COVID-19 pandemic, small sample size of adolescents with longitudinal follow-up, and the use of a cohort study with a larger proportion of young adults to evaluate a media campaign that was targeted to adolescents. There was a greater proportion of 12-17 year-old females in the PACE Vermont study compared to state estimates (Vermont Department of Health, 2021a, 2021b), though the study had representation across all counties, and substance use prevalence estimates for both age groups (12-17 and 18-25) were comparable to state-level estimates from the National Survey on Drug Use and Health (Substance Abuse and Mental Health Services Administration, 2022; Villanti et al., 2020). Susceptibility items were only assessed at Wave 9, so we could not account for baseline susceptibility to use EVPs. Last, although we assessed awareness of the Unhyped brand, we could not confirm direct exposure to the campaign; furthermore, recall of vaping phrases from the campaign (presented in Table 4) were only assessed among people reporting awareness of the campaign, so we may be missing recall of this information among people who were unaware that they were exposed to the campaign.

Recent FDA campaigns have used targeted digital media to disseminate tobacco prevention messages to at-risk groups, including hip hop youth (Fresh Empire) and LGBT young adults (This Free Life) (Guillory et al., 2021; Guillory et al., 2020). These national campaigns have reported impact on reach, awareness, and receptivity, with two showing modest effects on campaign-targeted beliefs (Crankshaw et al., 2022; Guillory et al., 2022). Findings from the current study suggest the short-term effectiveness of state-level targeted digital media vaping prevention efforts on campaign-targeted harm perceptions and use susceptibility. Differences in brand awareness by sociodemographic characteristics highlight the potential for campaign targeting to reduce vaping in young people at risk for tobacco use, including young adults and those who have tried substances (Dai et al., 2021; Wiggins et al., 2020). Our findings relevant to Vermont, where 61% of the population is considered rural (U.S. Census Bureau, 2012), are particularly important given rapid increases in EVP use among rural youth (Dai et al., 2021). Additionally, our study suggests that these campaigns may also reduce susceptibility to vaping, underscoring the need for studies that are adequately powered to assess campaign effects on vaping use behaviors. Despite these promising effects, we did not find any campaign effects on harm perceptions of EVPs relative to other products, lifetime substance use, current EVP use, or attempts to quit EVPs.

Our findings suggest the effectiveness of Vermont's targeted digital vaping prevention campaign on short-term prevention and longer-term susceptibility in the context of these efforts and other policy measures to reduce vaping in young people. They also underscore the need to conduct public health counter-marketing to reduce the impact of exposure to vape products on social media (Lee et al., 2021). Future research is needed to demonstrate the effectiveness of state efforts using targeted digital vaping prevention campaigns on vaping behaviors.

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Declaration of interest

The authors report there are no competing interests to declare.

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Data availability statement

The data underlying this article cannot be shared publicly due to the need to protect the privacy of individuals that participated in the study. Anonymized data will be shared on reasonable request to the corresponding author with permission of the University of Vermont and Vermont Department of Health. Individuals requesting a copy of study data will need to submit a detailed research plan that includes the purpose of the proposed research, the variables required, the duration of the analysis phase, IRB approval with FWA information, investigator training in human subjects and other approvals specific to the individual datasets. They will also be required to submit a Data Protection Plan or Institutional Privacy Policy, describing the computing environment in which data will be managed and analyzed and how physical access to computing equipment will be controlled. Data will only be released once all IRB approvals and Human Subjects concerns have been addressed. Data collection instruments, codebooks, and other data documentation will be made available in conjunction with this dataset in a standard format that is readable across a variety of applications and operating system platforms. The PI will be required to participate as investigators on any project requiring data sharing. Analytic code is available upon request.

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